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Release date 2023-5-31			file version	V1.7

TXW806 data sheet



	Zhuhai Taixin Semiconductor Co., Ltd. Zhuhai Taixin Semiconductor Co., Limited	Zhuhai High-tech Zone Harbor No. 1 Science and Technology Innovation Park Port Building 11, 3rd floor
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Revision history

date version		describe	Revised by
2023-5-31	V1.7	adds TXW806-851D and low power consumption related performance descriptions	TX
2022-12-12	V1.6	adds TXW806-900 QFN56 description	
2022-08-04	V1.5	adds VCAM LDO load capacity indicator	TX
2022-06-22	V1.4	calibrated RF performance indicators	TX
2022-05-28	V1.3	Added TXW806-G00 QFN48 pin diagram Added PB0~5 QSPI function description	TX
2022-04-29	V1.2	optimized QFN48 pin diagram	TX
2022-04-02	V1.1	supplements most electrical parameter performance indicators	TX
2022-02-17	V1.0	Initial version, there will be updates in the future without further notice, please contact Our sales staff obtain the latest version	TX

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1. Product Overview

1.1. Description

TXW806 is a low-power, high-performance, highly integrated 2.4GHz Wi-Fi+ small wireless multi-mode IoT SOC chip.

Integrated IEEE 802.11 b/g/n baseband and RF (Radio Frequency) circuits, including power amplifier PA (Power

Amplifier), low noise amplifier LNA (Low Noise Amplifier), RF balun, antenna switch and electrical

Source management module, etc.

The TXW806 Wi-Fi baseband implements Orthogonal Frequency Division Multiplexing (OFDM) technology and is backward compatible with Direct Sequence Spread Spectrum (DSSS),

Complementary code keying (CCK) technology, supporting IEEE 802.11 b/g/n protocol. Supports 20MHz standard bandwidth and 5MHz/10MHz

Narrow bandwidth, providing a maximum physical layer rate of 72.2Mbit/s.

The TXW806 chip integrates a high-performance 32bit microprocessor, built-in MJPEG (supports VGA/720P), and provides DVP and USB2.0

High Speed Host/Device SDMMC Host/SDIO2.0 Slave/RS485 MAC/ SPI Master & Device

Rich peripheral interfaces such as UART, IIC, IIS, PDM, IR Send/Receive, PWM, GPIO and ADC/DAC support

Keep running programs on SPI Flash. Supports RTOS and third-party components, and provides open and easy-to-use development and debugging

test environment.

The TXW806 series includes multiple models, providing QFN56, QFN48, QFN40 and other mainstream packaging forms; according to different

The packaging form and peripheral resource configuration in the device are different; some packages support built-in PSRAM and Flash.

Application occasions:

• Wireless audio and video applications

• Visual otoscope, aerial photography

• Endoscopic

• MiniDV

• Driving recorder

• Low-power audio and video products (video doorbells, etc.)

1.2. Features

• Wi-Fi MAC & PHY

- Support IEEE 802.11 b/g/n specification
- Support 1T1R mode, data rate up to 72.2Mbps
- Excellent transmit power and receiving sensitivity
- Support Bluetooth fast network configuration
- Built-in PA, LNA and RF switches
- Support STA, AP, AP+STA functions
- Support relay
- Frame aggregation (TX/RX A-MPDU, RX A-MSDU)
- Support WPA/WPA2/WPA3

• MCU

- CK803 CPU, maximum frequency 180Mhz
- 294KB SRAM
- Supports external multiple frequency crystal oscillators and supports sharing crystal oscillator input with the system main control chip
- Support RTC
- Support 32KHz crystal oscillator
- 22 timers
 - 2 normal power timers
 - 4 low power mode timers
 - 6 32bit timers, supporting infrared transceiver and LED strip driver
 - 8 16bit timers
 - 1 24bit system tick timer
 - 1 RTCC timer

• Supports ULP and LP low power consumption modes, ULP mode current <6uA@25°C, supports multiple IO wake-ups at the same time

• Built-in temperature sensor

• Built-in LVD detection

• Built-in watchdog

• Built-in DC-DC power management module

• Built-in LDO output, output voltage range 1.8~3.3V

• 88-bit chip unique ID (UID)

• Peripherals

• 40 programmable GPIOs, supporting edge or level triggered interrupts

• 2 channels of 12bit ADC, can be reused as DAC

• 1 channel QSPI, supports external SPI FLASH or PSRAM

• 1 channel CMOS Sensor 8-bit DVP, the interface supports up to 96Mhz

• 1 channel Motion JPEG encoder

• 2-way IIS/PCM

• 1 channel PDM

• 1 • SDIO2.0 High Speed Device

• 1 channel SD Host Controller

• 1 • USB2.0 High Speed Device/Host

• 5-way SPI interface Master/Slave (2 of which can be configured as IIC Master/Slave)

• 4 UART interfaces (2 of which support flow control and RS485)

• 2 channels of infrared transmission and multi-channel reception

• 18 channels of PWM output (multiplex timer), including 4 channels of independent 12-bit PWM, supporting low power consumption mode PWM,

Supports 3 pairs of complementary PWM;

• 1 channel 8COM-12SEG LED driver

• 1-way touch button detection

- 1 channel RMI Ethernet controller

- Start interface

- SDIO2.0 Device
 - USB2.0 Device
 - SPI Slave
 - UART

- SPI FLASH

- Data security

- Support AES 128/192/256 encryption and decryption

- Support 5/7/8/16/32 bit CRC verification

- Support SPI Flash firmware encryption protection

- TRNG true random number generator

- Package

- QFN56 6x6 package

- QFN48/QFN40 5x5 package

- Temperature range

- -40°C to 85°C

1.3. Functional block diagram



Nor Flash and PSRAM are built into some chip packages.

1.4. Pin assignment

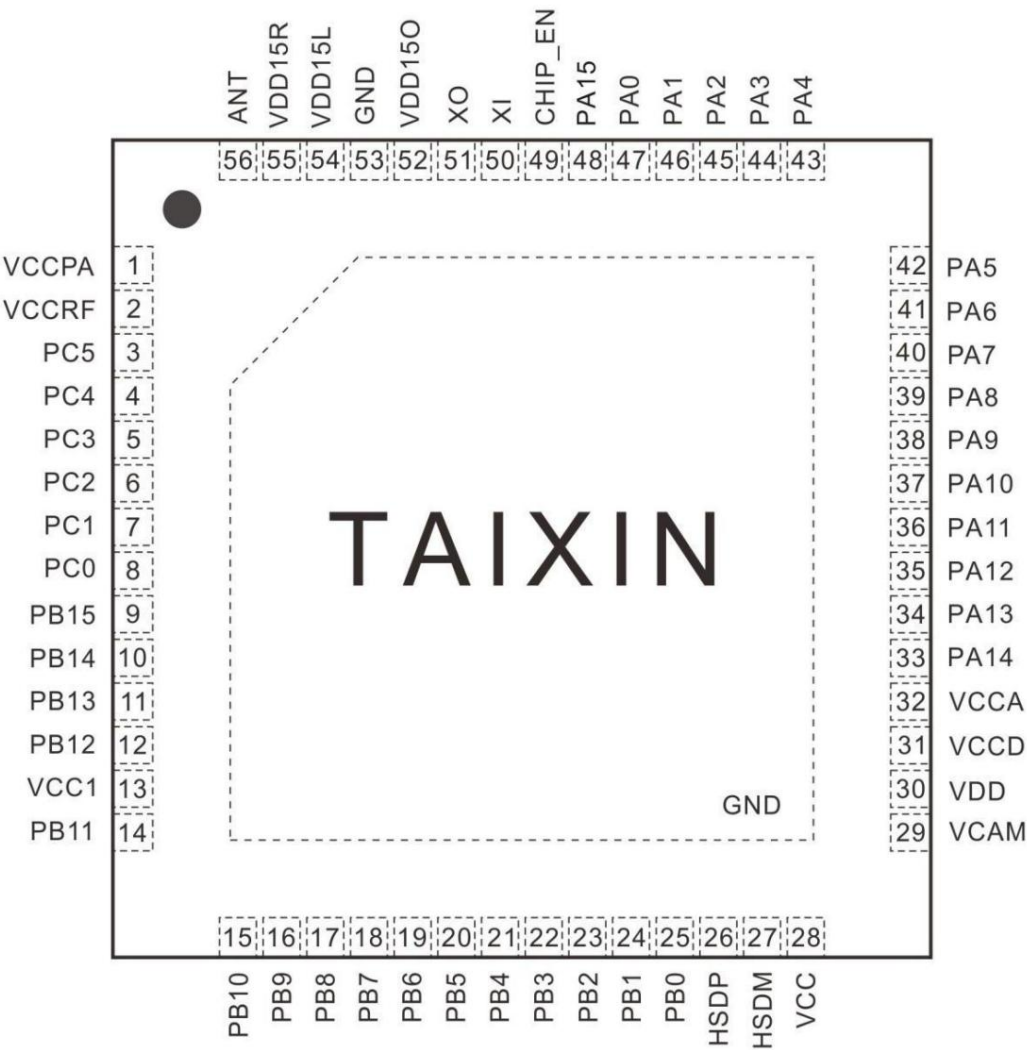


Figure 1-4-1 TXW806-9xx QFN56 package pin diagram

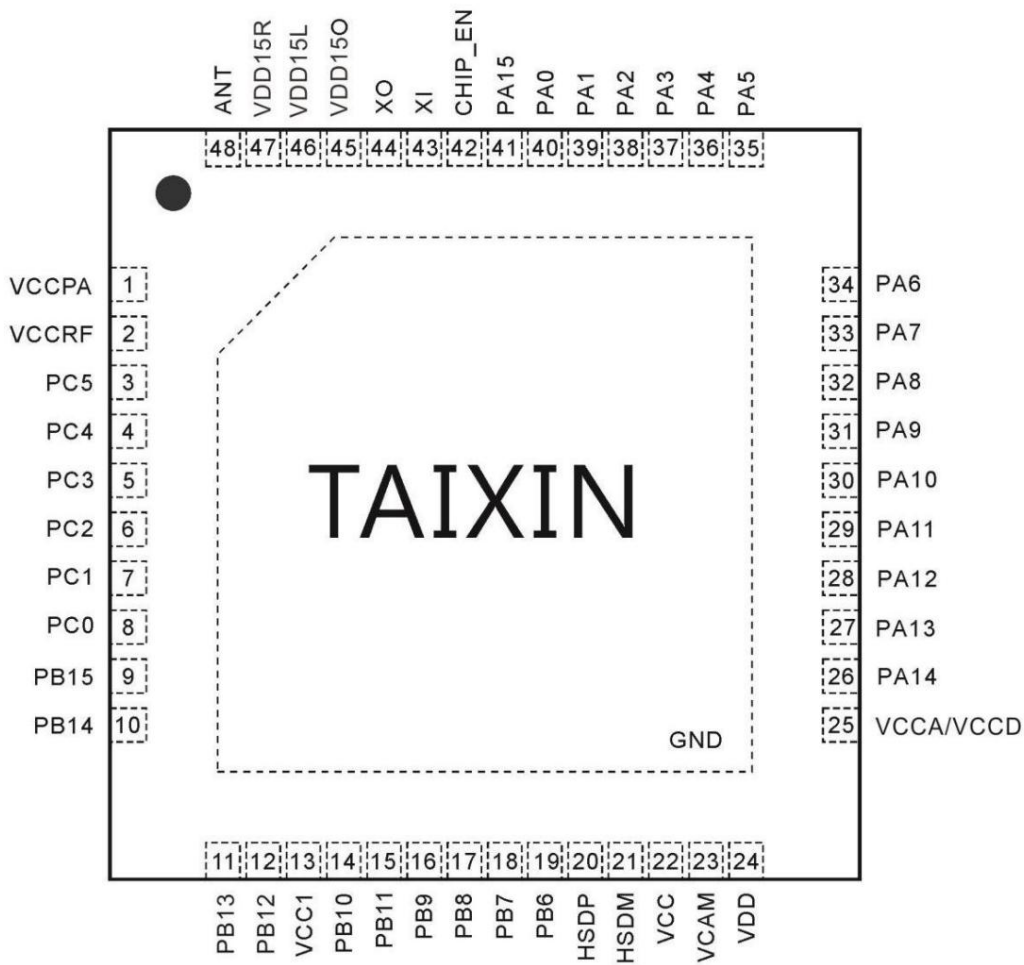


Figure 1-4-2 TXW806-8xx QFN48 package pin diagram

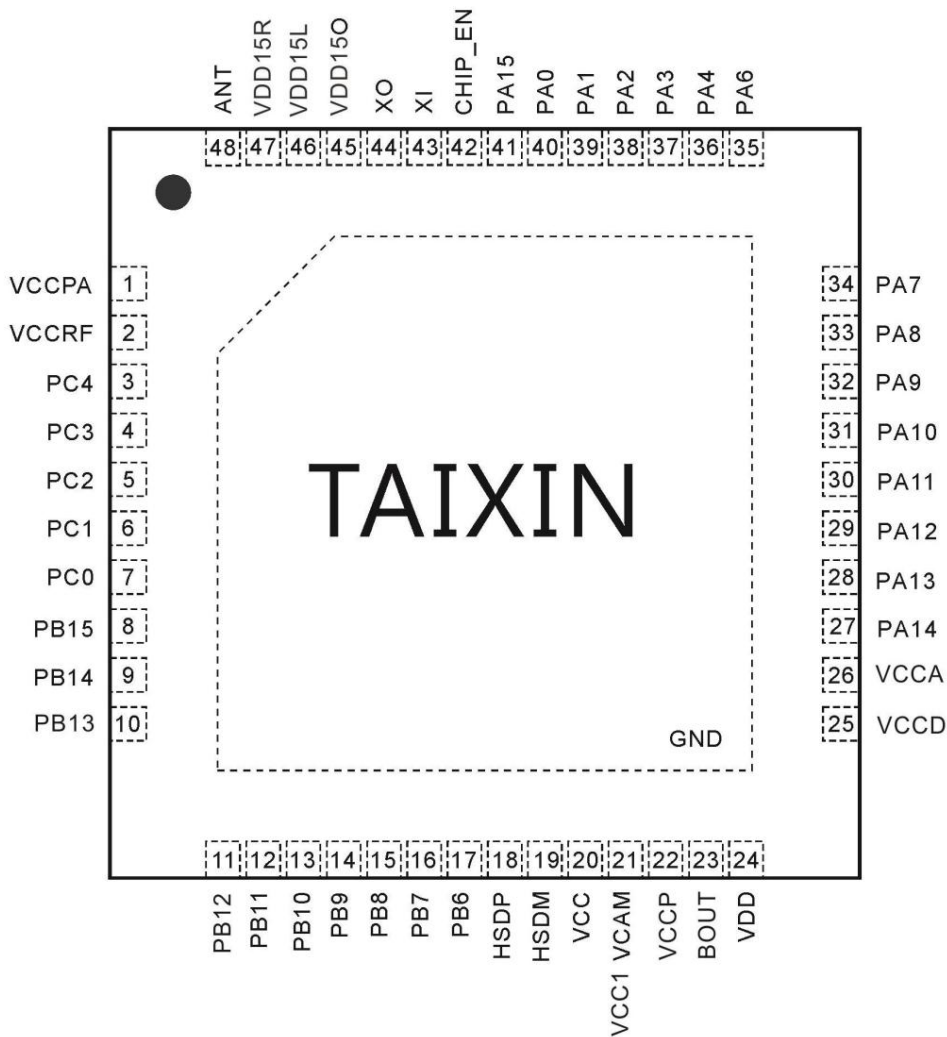


Figure 1-4-3 TXW806-8xxD QFN48 package pin diagram

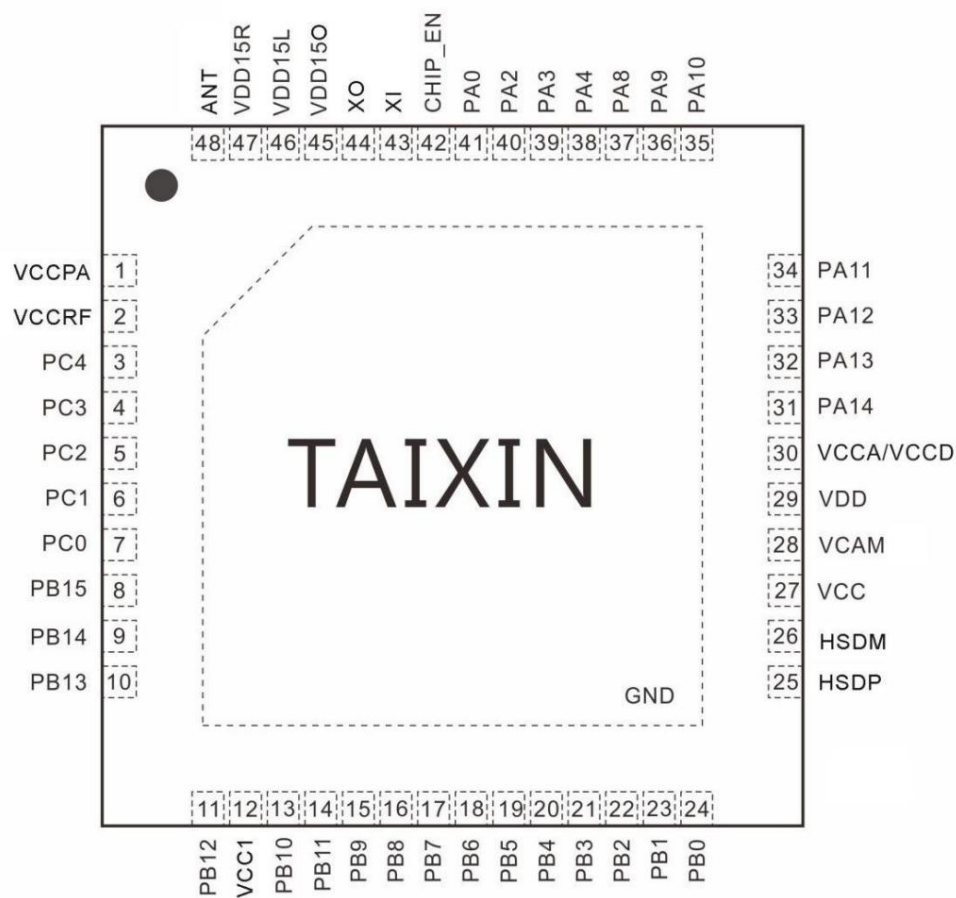


Figure 1-4-4 TXW806-Gxx QFN48 package pin diagram

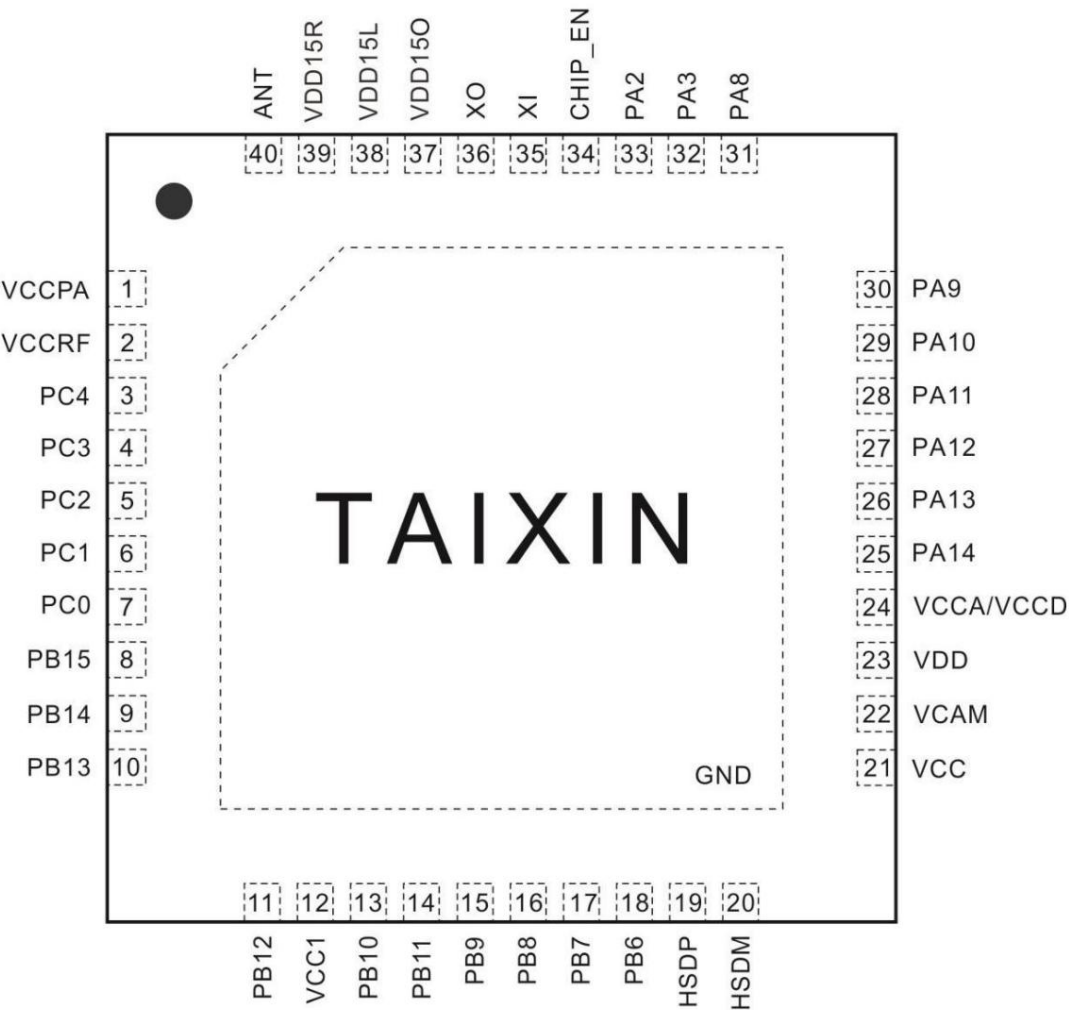


Figure 1-4-4 TXW806-6xx QFN40 package pin diagram

1.5. Packaging information

The models of TXW806 series are as follows:

Table 1-1 Packaging information

model	encapsulation	size	Package
TXW806-9xx	QFN56	6x6	
TXW806-8xx	QFN48	5x5	
TXW806-Gxx			
TXW806-6xx	QFN40	5x5	

1.6. Package size diagram

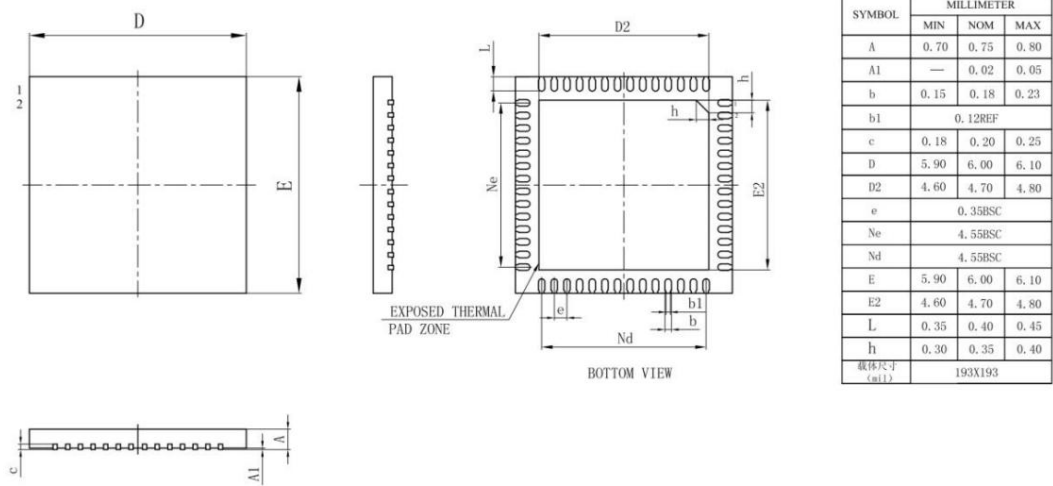


Figure 1-6-1 QFN56 package POD diagram

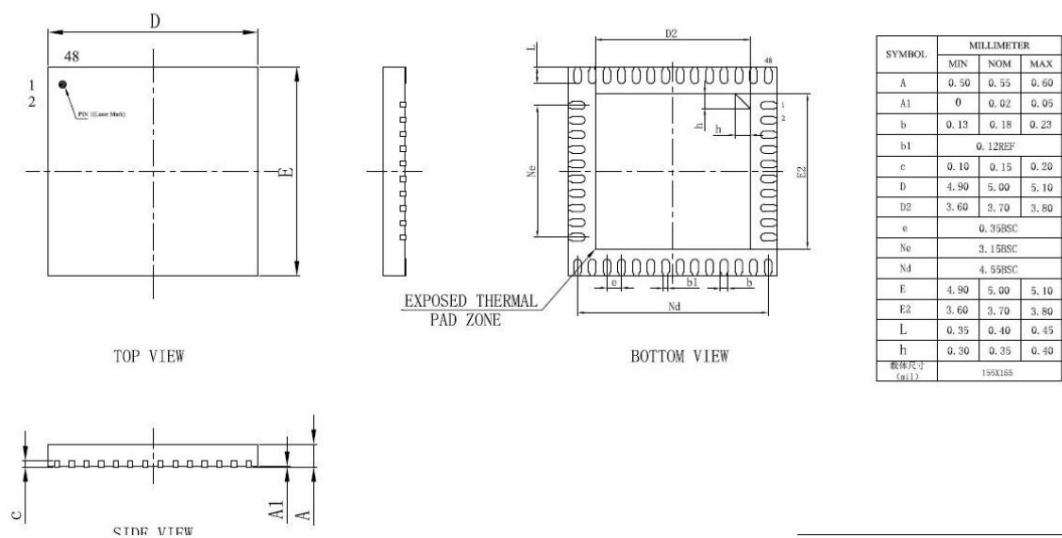


Figure 1-6-2 QFN48 package POD diagram

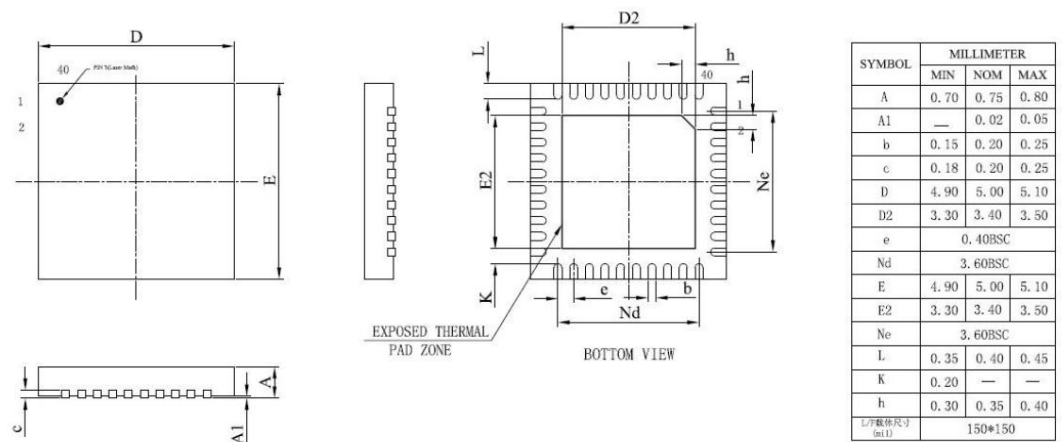


Figure 1-6-3 QFN40 package POD diagram

1.7. Pin description

1.7.1. Pin specific functions

Table 1-7-1 GPIO pin specific function table

Pin name	I/O function description	Reset state	multiplexing function
VCCPA	A power supply	.	none
VCCRF	A power supply	.	none
VCC1	A power supply	.	none
VCC	A power supply	.	none
VCAM	A power supply	.	none
VCCA	A power supply	.	none
VCCD	A power supply	.	none
VDD	A power supply	.	none
VDD15O	A power supply	.	none
VDD15L	A power supply	.	none
VDD15R	A power supply	.	none
BOUT	A	.	none
XI	A crystal oscillator input	.	none
XO	A crystal oscillator output	.	none
HSDP	A	USB2.0 DP	Can be reused as GPIO PC6
HSDM	A	USB2.0 DM	Can be reused as GPIO PC7
ON	A RF antenna	.	none
GND	A	ePAD ground wire	none
CHIP_EN	I chip enable: 0: Chip off 1: Chip enable	Input pull-up	None

PA0	I/O Each IO port can be set Set as input or output model Input modules can be enabled Internal pull-up The output module can be configured Open drain output	Input high impedance	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK0 [Touch button external channel 0] LED_COM3 [LED COM3]
PA1		Input high impedance	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK1 [Touch button external channel 1] LED_COM2 [LED COM2]
PA2		Input high impedance	LEDTMR0_PWM_OUT ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK2 [Touch button external channel 2] LED_COM1 [LED COM1] QSPI_IO0 [QSPI IO0]
PA3		Input high impedance	LEDTMR1_PWM_OUT ADKEY0_N0 [ADC0 external channel N0] ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK3 [Touch button external channel 3] LED_COM0 [LED COM0] QSPI_IO3 [QSPI IO3]
PA4		Input high impedance	ADKEY0_N1 [ADC0 external channel N1] ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK4 [Touch button external channel 4] LED_SEG0 [LED SEG0]
PA5		Input high impedance	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK5 [Touch button external channel 5] LED_SEG1 [LED SEG1]
PA6		Input high impedance	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 [ADC1 external channel P0] TK6 [Touch button external channel 6] LED_SEG2 [LED SEG2]

PA7			Input high impedance	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 ADC1 external channel P0 TK7 [Touch button external channel 7] LED_SEG3 LED SEG3
PA8	I/O		Input high impedance	LEDTMR2_PWM_OUT ADKEY1_N0 ADC1 external channel N0 ADKEY0_P1 ADC0 external channel P1 ADKEY1_P0 ADC1 external channel P0 TK8 [Touch button external channel 8] LED_SEG4 LED SEG4 QSPI_CLK QSPI CLK
PA9	I/O		Input pull-up	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 ADC1 external channel P0 TK9 [Touch button external channel 9] LED_SEG9 LED SEG9 QSPI_IO2 QSPI IO2
PA10	I/O		Input pull-up	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0 ADC1 external channel P0 TK10 [Touch button external channel 10] LED_SEG8 LED SEG8 QSPI_IO1 QSPI IO1
PA11			Input high impedance	LEDTMR3_PWM_OUT ADKEY1_P1 ADC1 external channel P1 ADKEY0_P1 ADC0 external channel P1 ADKEY1_P0 ADC1 external channel P0 TK11 [Touch button external channel 11] LED_SEG5 LED SEG5 QSPI_IO2 QSPI IO2
PA12			Input high impedance	LXOSCO [low speed crystal oscillator output] ADKEY1_P2 ADC1 external channel P2 ADKEY0_P1 ADC0 external channel P1 ADKEY1_P0 ADC1 external channel P0 TK12 [Touch button external channel 12] LED_SEG10 LED SEG10

				QSPI_IO1yQSPI IO1y
PA13			Input high-impedance	LXOSCI [low-speed crystal oscillator input] ADKEY1_N1yADC1 external channel N1y ADKEY0_P1yADC0 external channel P1y ADKEY1_P0yADC1 external channel P0y TK13 [Touch button external channel 13] LED_SEG6yLED SEG6y QSPI_IO0yQSPI IO0y
PA14	I/O		Input high impedance	TK_CAP [touch button external capacitor interface] ADKEY1_P3yADC1 external channel P3y ADKEY0_P1yADC0 external channel P1y ADKEY1_P0yADC1 external channel P0y LED_SEG7yLED SEG7y QSPI_IO3yQSPI IO3y
PA15			Input high impedance	ADKEY0_P1 [ADC0 external channel P1] ADKEY1_P0yADC1 external channel P0y LED_SEG11yLED SEG11y LED_COM4yLED COM4y MCLR
PB0	I/O		Input high impedance	SD_DAT1ySDIO DAT1y QSPI_IO0yQSPI IO0y
PB1	I/O		Input high impedance	SD_DAT0ySDIO DAT0y QSPI_CLKyQSPI CLKy
PB2	I/O		Input high impedance	SD_CLKySDIO CLKy QSPI_IO3yQSPI IO3y
PB3	I/O		Input high impedance	SD_CMD [SDIO CMD] QSPI_IO2yQSPI IO2y
PB4	I/O		Input high impedance	SD_DAT3 [SDIO DAT3] QSPI_CSyQSPI CSy
PB5	I/O		Input high impedance	SD_DAT2ySDIO DAT2y

				QSPI_IO1yQSPI IO1y
PB6	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK16 [Touch button external channel 16] SD_DAT1ySDIO DAT1y DVP_VSYNCyDVP interface VSYNCy
PB7	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK17 [Touch button external channel 17] SD_DAT0ySDIO DAT0y DVP_HSYNCyDVP interface HSYNCy
PB8	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK18 [Touch button external channel 18] SD_CLKySDIO CLKy DVP_DATA_IN7 [DVP access DAT IN7]
PB9	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK19 [Touch button external channel 19] SD_CMDySDIO CMDy DVP_MCLKyDVP interface MCLKy
PB10	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK20 [Touch button external channel 20] SD_DAT3ySDIO DAT3y DVP_DATA_IN6 [DVP access DAT IN6]
PB11	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y

				TK21 [Touch button external channel 21] SD_DAT2ySDIO DAT2y DVP_DATA_IN5yDVP interface DAT IN5y
PB12	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK22 [Touch button external channel 22] RMII_RXD0yEthernet RMII RXD0y DVP_PIXEL_CLK_INyDVP interface pixel clocky
PB13	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK23 [Touch button external channel 23] RMII_RXD1yEthernet RMII RXD1y DVP_DATA_IN4 [DVP access DAT IN4]
PB14	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK24 [Touch button external channel 24] RMII_REF_CLKINyEthernet RMII REFCLKy DVP_DATA_IN0 [DVP access DAT IN0]
PB15	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK25 [Touch button external channel 25] RMII_TXD0yEthernet RMII TXD0y DVP_DATA_IN3 [DVP access DAT IN3]
PC0	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0yADC0 external channel P0y ADKEY1_N2yADC1 external channel N2y TK26 [Touch button external channel 26] RMII_TXD1yEthernet RMII TXD1y DVP_DATA_IN1 [DVP access DAT IN1]

PC1	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0 [ADC0 external channel P0] ADKEY1_N2 [ADC1 external channel N2] TK27 [Touch button external channel 27] RMII_TX_EN [Ethernet RMII TXEN] DVP_DATA_IN2 [DVP access DAT IN2]
PC2	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0 [ADC0 external channel P0] ADKEY1_N2 [ADC1 external channel N2] TK28 [Touch button external channel 28] RMII_CRS_DV [Ethernet RMII CRS_DV] LED_COM4 [LED COM4]
PC3	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0 [ADC0 external channel P0] ADKEY1_N2 [ADC1 external channel N2] TK29 [Touch button external channel 29] RMII_RX_ER [Ethernet RMII RX_ER] LED_COM5 [LED COM5]
PC4	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0 [ADC0 external channel P0] ADKEY1_N2 [ADC1 external channel N2] TK30 [Touch button external channel 30] LED_SEG9 [LED SEG9] LED_COM6 [LED COM6]
PC5	I/O		Input high impedance	ADKEY0_N2 [ADC0 external channel N2] ADKEY0_P0 [ADC0 external channel P0] ADKEY1_N2 [ADC1 external channel N2] TK31 [Touch button external channel 31] LED_SEG8 [LED SEG8] LED_COM7 [LED COM7]

NOTE: The other digital functions of all pins can be arbitrarily selected from the following function mappings for flexible configuration.

1.7.2. Pin output arbitrary mapping function

Table 1-7-2 GPIO output function arbitrary mapping table

function number	function name	Function Description
1	COMP_DOUT_DIG0	Comparator 0 digital IO output
2	COMP_DOUT_DIG1	Comparator 1 digital IO output
3	GRANT_BLE_SWITCH_O	Bluetooth coexistence SWITCH signal
4	GRANT_BLE	Bluetooth coexistence BLE arbitration signal
5	GRANT_WI-FI_SWITCH_O	Bluetooth coexistence WI-FI SWITCH signal
6	RF_SWITCH_EN1	External RF switch enable 1
7	RF_SWITCH_EN0	External RF switch enable 0
8	ANTENNA_SEL	Dual antenna selection signal
9	PA_EN	External PA enable signal
10	RF_EXT_LNA_EN	External RF LNA enable signal
11	RF_TX_EN_FEM	External RF FEM transmit enable
12	RF_RX_EN_FEM	External RF FEM receive enable
13	UART4_TX	UART4 TX output
14	UART5_TX	UART5 TX output
15	UART1_RTS_RE_O	UART1 RTS RE output
16	UART1_CTS_DE_OUT	UART1 RTS DE output
17	UART1_OUT	UART1 TX output
18	UART0_RTS_RE_O	UART0 RTS RE output
19	UART0_CTS_DE_OUT	UART0 CTS DE output
20	UART0_OUT	UART0 TX output
21	STMR3_PWM_OUT	SIMPLE TIMER3 PWM output
22	STMR2_PWM_OUT	SIMPLE TIMER2 PWM output
23	STMR1_PWM_OUT	SIMPLE TIMER1 PWM output
24	STMR0_PWM_OUT	SIMPLE TIMER0 PWM output
25	SUPTMR5_PWMOUT	Advanced TIMER5 PWM output
26	SUPTMR4_PWMOUT	Advanced TIMER4 PWM output
27	SUPTMR3_PWMOUT	Advanced TIMER3 PWM output
28	SUPTMR2_PWMOUT	Advanced TIMER2 PWM output
29	SUPTMR1_PWMOUT	Advanced TIMER1 PWM output
30	SUPTMR0_PWMOUT	Advanced TIMER0 PWM output

31	SDHOST_SCLK_O	SDC HOST SDCLK output
32	TMR3_PWM_OUT	TIMER3 PWM output
33	TMR2_PWM_OUT	TIMER2 PWM output
34	TMR1_PWM_OUT	TIMER1 PWM output
35	TMR0_PWM_OUT	TIMER0 PWM output
36	LED_TMR0_PWM_OUT	LED TIMER0 PWM output
37	LED_TMR1_PWM_OUT	LED TIMER1 PWM output
38	LED_TMR2_PWM_OUT	LED TIMER2 PWM output
39	LED_TMR3_PWM_OUT	LED TIMER3 PWM output
40	PDM_MCLK	PDM MCLK output
41	QSPI_NSS1_OUT	QSPI chip select 1 output
42	SPI0_NSS_OUT	SPI0 chip select output
43	SPI0_SCK_OUT	SPI0 CLK output
44	SPI0_IO0_OUT	SPI0 IO0 output
45	SPI0_IO1_OUT	SPI0 IO1 output
46	SPI0_IO2_OUT	SPI0 IO2 output
47	SPI0_IO3_OUT	SPI0 IO3 output
48	SPI1_NSS_OUT	SPI1 chip select output
49	SPI1_SCK_OUT	SPI1 CLK output
50	SPI1_IO0_OUT	SPI1 IO0 output
51	SPI1_IO1_OUT	SPI1 IO1 output
52	SPI1_IO2_OUT	SPI1 IO2 output
53	SPI1_IO3_OUT	SPI1 IO3 output
54	SPI2_NSS_OUT	SPI2 chip select output
55	SPI2_SCK_OUT	SPI2 CLK output
56	SPI2_IO0_OUT	SPI2 IO0 output
57	SPI2_IO1_OUT	SPI2 IO1 output
58	SPI2_IO2_OUT	SPI2 IO2 output
59	SPI2_IO3_OUT	SPI2 IO3 output
60	SPI5_SCK_OUT	SPI5 CLK output
61	SPI5_IO0_OUT	SPI5 IO0 output
62	SPI5_IO1_OUT	SPI5 IO1 output
63	SPI6_SCK_OUT	SPI6 CLK output
64	SPI6_IO0_OUT	SPI6 IO0 output

65	SPI6_IO1_OUT	SPI6 IO1 output
66	SDHOST_CMD_OUT	SDHOST CMD output
67	SDHOST_DAT0_OUT	SDHOST DAT0 output
68	SDHOST_DAT1_OUT	SDHOST DAT1 output
69	SDHOST_DAT2_OUT	SDHOST DAT2 output
70	SDHOST_DAT3_OUT	SDHOST DAT3 output
71	IIS0_MCLK_OUT	IIS0 MCLK output
72	IIS0_WSCLK_OUT	IIS0 WS output
73	IIS0_BCLK_OUT	IIS0 BCLK output
74	IIS0_DO	IIS0 DAT output
75	IIS1_MCLK_OUT	IIS1 MCLK output
76	IIS1_WSCLK_OUT	IIS1 WS output
77	IIS1_BCLK_OUT	IIS1 BCLK output
78	IIS1_DO	IIS1 DAT output
79	CLK_TO_IO	Clock source IO output

1.7.3. Pin input arbitrary mapping function

Table 1-7-3 GPIO input function arbitrary mapping table

function number	function name	Function Description
1	TMR0_CAP_IN	TIMER0 capture input
2	TMR0_SYNC_IN	TIMER0 synchronous input
3	TMR1_CAP_IN	TIMER1 capture input
4	TMR2_CAP_IN	TIMER2 capture input
5	TMR3_CAP_IN	TIMER3 capture input
6	PDM_DATA_IN	PDM DATA input
7	PTA_REQ_IN	PTA REQ input
8	PTA_PRI_IN	PTA PRI input
9	FREQ_IND_IN	FREQ IND input
10	STMR0_CAP_IN	Simple TIMER0 capture input
11	STMR1_CAP_IN	Simple TIMER1 capture input
12	STMR2_CAP_IN	Simple TIMER2 capture input

13	STMR3_CAP_IN	Simple TIMER3 capture input
14	PORT_WKUP_IN0	IO wake-up channel 0 input
15	PORT_WKUP_IN1	IO wake-up channel 1 input
16	PORT_WKUP_IN2	IO wake-up channel 2 input
17	PORT_WKUP_IN3	IO wake-up channel 3 input
18	UART0_IN	UART0 RX input
19	UART0_CTS_DE_IN	UART0 CTS/DE input
20	UART1_IN	UART1 RX input
21	UART1_CTS_DE_IN	UART1 CTS/DE input
22	FB_IN/EXT_PA/SYS_NMI	FB_IN/EXT_PA/SYS_NMI input
23	UART4_IN	UART4 RX input
24	UART5_IN	UART5 RX input
25	SPI0_NSS_IN	SPI0 NSS input
26	SPI0_SCK_IN	SPI0 SCK input
27	SPI0_IO0_IN	SPI0 IO0 input
28	SPI0_IO1_IN	SPI0 IO1 input
29	SPI0_IO2_IN	SPI0 IO2 input
30	SPI0_IO3_IN	SPI0 IO3 input
31	SPI1_NSS_IN	SPI1 NSS input
32	SPI1_SCK_IN	SPI1 SCK input
33	SPI1_IO0_IN	SPI1 IO0 input
34	SPI1_IO1_IN	SPI1 IO1 input
35	SPI1_IO2_IN	SPI1 IO2 input
36	SPI1_IO3_IN	SPI1 IO3 input
37	SPI2_NSS_IN	SPI2 NSS input
38	SPI2_SCK_IN	SPI2 SCK input
39	SPI2_IO0_IN	SPI2 IO0 input
40	SPI2_IO1_IN	SPI2 IO1 input
41	SPI2_IO2_IN	SPI2 IO2 input
42	SPI2_IO3_IN	SPI2 IO3 input
43	SPI5_SCK_IN	SPI5 SCK input
44	SPI5_IO0_IN	SPI5 IO0 input
45	SPI5_IO1_IN	SPI5 IO1 input
46	SPI6_SCK_IN	SPI6 SCK input

47	SPI6_IO0_IN	SPI6 IO0 input
48	SPI6_IO1_IN	SPI6 IO1 input
49	SDHOST_CMD_IN	SDHOST CMD input
50	SDHOST_DAT0_IN	SDHOST DAT0 input
51	SDHOST_DAT1_IN	SDHOST DAT1 input
52	SDHOST_DAT2_IN	SDHOST DAT2 input
53	SDHOST_DAT3_IN	SDHOST DAT3 input
54	IIS0_MCLK_IN	IIS0 MCLK input
55	IIS0_WSCLK_IN	IIS0 WS input
56	IIS0_BCLK_IN	IIS0 BCLK input
57	IIS0_DI	IIS0 DAT input
58	IIS1_MCLK_IN	IIS1 MCLK input
59	IIS1_WSCLK_IN	IIS1 WS input
60	IIS1_BCLK_IN	IIS1 BCLK input
61	IIS1_DI	IIS1 DAT input

2. Function description

2.1. Processor and memory

2.1.1. CPU

The TXW806 series chips are equipped with the CK803 processor, with a maximum frequency of 180MHz and the following features:

- Reduced instruction set processor architecture (RISC)
- 32-bit data, 16-bit/32-bit mixed encoding instructions
- 3-stage assembly line
- 32KByte cache
- Single-cycle fast hardware multiplier
- Vectored interrupt controller and tick timer
- Interrupt response latency is only 13 processor cycles

2.1.2. Memory

The on-chip memory of the TXW806 series chips includes:

- ROM: BOOTLOADER and some kernel functions
- 294KB SRAM: data and instruction space
- 104Bit EFUSE: used for keys and customer customization

2.1.3. Memory expansion

The TXW806 series of chips supports memory expansion, SPI Flash and PSRAM that expand the SPI interface (some chips already built in). Supports external QSPI Flash and PSRAM through on-chip QSPI controller (supports up to two memory expansions server), supports XIP function. QSPI supports external memory access space mapping of up to 32MByte of Flash and PSRAM.

2.2. System clock

The clock sources of TXW806 series chips include:

- 128KHz LIRC

- 10MHz HIRC

- 32.768KHz low-speed crystal oscillator

- 26~40MHz high-speed crystal oscillator

- 360MHz fractional frequency division system PLL

- 480MHz fractional frequency USB2.0 PLL

- External IO input clock source

2.3. Analog peripherals

2.3.1. Analog-to-digital converter (SARADC)

The TXW806 series chip integrates two 12bit SARADCs and can work in three functional modes: ADC/DAC/CMP.

Formula, the specific characteristics are as follows:

- Supports clock input up to 1MHz

- Up to 62.5Ksps

- Support analog hysteresis comparison function

- Support 12bit ADC/DAC conversion accuracy

- Support timer0/1/2/3/4/5/6 and software trigger ADC for conversion

2.3.2. Temperature sensor

The TXW806 series chip integrates a temperature sensor, which is connected to the temperature sensor through an internal channel of the ADC.

The voltage sampling is used to calculate the temperature value inside the chip.

2.3.3. Touch sensor

The TXW806 series chip integrates a capacitive touch sensor and can support up to 30 touch buttons. Tool

The body characteristics are as follows:

- All 30 IOs support touch detection
- Supports touch detection that automatically scans all enabled IOs
- Supports manual channel-by-channel scanning of all enabled IO touch detection
- Supports continuous multiple conversions of a single touch detection IO
- Support waking up the CPU/system
- Supports single touch detection IO and generates an interrupt after a single conversion is completed
- Supports single touch detection IO and generates interrupts after multiple consecutive conversions
- Supports interrupt generation after touch detection of all enabled IOs
- The result of each conversion supports DMA to RAM
- Supports parallel multiple touch detection IOs to be connected to the analog module for detection at the same time

2.3.4. USB2.0

The TXW806 series chips have built-in USB2.0 Controller and USB2.0 PHY, and are compatible with standard USB2.0 High/Full

Speed Host & Device Protocol.

The TXW806 series chip has high-performance USB2.0 high-speed PHY and self-developed innovative architecture CDR patented technology, which can ensure

It can also work normally in harsh environments. No need to plug in an independent USB2.0 PHY to support high-speed mode, theoretical speed

Up to 480Mbps. Support crystal-free USB. When the USB2.0 function is not needed in the application solution, it can be used as two

GPIO to use.

2.3.5. XOSC

The TXW806 series chip integrates a high-speed crystal oscillator circuit module and requires an external 26~40MHz passive crystal oscillator.

2.3.6. PLL

The TXW806 series chips integrate 2 high-performance fractional frequency PLLs, which are 360MHz system fractional frequency PLL, and 480MHz USB2.0 PHY fractional PLL.

2.4. Digital peripherals

2.4.1. GPIO

The TXW806 series chips integrate up to 40 general-purpose GPIOs. The specific features are as follows:

- The pull-up and pull-down resistors can be configured, and the configured resistance values are 4.7K Ω , 10K Ω , and 100K Ω ;
- Supports 16-level IO output driver configurability, the configuration range is 4mA~64mA, each gear is 4mA;
- Supports configurable switch for IO input hysteresis function;
- Support open-drain output high and low functions;
- Supports IO status latch function after power outage in low power consumption mode;
- Support ADC analog input and TouchKey analog input function;
- Support independent digital IO input and output direction enable, and turn off the input and output function of digital IO;

2.4.2. SPI

The TXW806 series chip integrates 5 SPIs, including 2 simple function SPIs (SPI5/6) and 3 complex function SPIs. capable SPI (SPI0/1/2).

SPI1/2 supports SPI and IIC composite functions, and its features are as follows:

SPI functional features are as follows:

- Support host mode and slave mode

- Support Motorola SPI

- Support master-slave half-duplex sending and receiving

- Serial clock with programmable polarity and phase

- Supports 4 modes

- Standard mode: SCLK, CS, IO0 (MOSI), IO1 (MISO), one SCLK transmits 1 bit

- 3-wire mode: SCLK, CS, IO0, one SCLK transmits 1 bit

- DUAL mode: SCLK, CS, IO0, IO1, one SCLK transmits 2 bits

- QUAL mode: SCLK, CS, IO0, IO1, IO2, IO3, one SCLK transmits 4 bits

- Transfer end flag with MCU interrupt

- Supports system clock with master mode baud rate up to 1/2

- Supports 1~32-bit data width optional

- Supports DMA(direct memory access)

- Supports sending high bit first or low bit first, optional

The functional characteristics of the IIC module are as follows:

- Support host mode and slave mode

- Support host clock synchronization and arbitration

- Supports the slave device to pull SCL low when the sending data is not ready or the receiving buffer is full.

- Support slave 7bit address or 10bit address

- Support DMA

SPI5/6 only supports simple function SPI, its features are as follows:

- Support two-line/three-line mode

- Support master-slave half-duplex sending and receiving

- Serial clock with programmable polarity and phase

- Transfer end flag with MCU interrupt

- Main mode supports communication rate up to 24Mbps

2.4.3. UART

The TXW806 series chip integrates 4 UARTs, including 2 simple function UARTs (UART4/5) and 2 complex

Functional UART (UART0/1).

The functional characteristics of UART0/1 are as follows:

- Support 8bit data and 9bit data mode

- Support parity check

- With 4 frames of data receiving buffer and one frame of data sending buffer

- Support receiving and sending DMA

- Support hardware flow control, RTS_N and CTS_N have enable signals respectively, and the receive buffer can be configured to have N (N=0,

RTS_N is valid after 1, 2, 3) data

- Hardware detection receiving time out, time out configuration range: 1~65536 bit rate time

- Support rs485 mode

The specific features of UART4/5 are as follows:

- Support half-duplex

- Support sending 9bit data

- Support software parity check

2.4.4. IIS_PCM

The TXW806 series chips integrate 2 IIS_PCM function modules for application extended audio functions. The specific features are:

The properties are as follows:

- Supports sending in IIS/left-aligned/right-aligned/PCM format

- Support left channel/right channel/stereo mode

- Support master/slave mode

- Supports 2-32bit working bit width

- Supports output MCLK mode and supports using input MCLK as the clock of master and slave.

- Support left and right channel interchange

- Support dual address automatic switching

2.4.5. PDM

The TXW806 series chip integrates a PDM function module for application extended audio functions.

The PDM module is an audio interface that handles digital MEMS microphones. It can provide a high-precision output clock and support

For left/right channel and stereo use, the chip integrates a PDM interface. The specific functions are as follows:

- Supports conversion of PDM data input by digital MEMS microphone into 16bit PCM data

- Support left channel/right channel/stereo mode

- Provides two decimation ratios of 64/128 PDM/PCM modes

- Support left and right channel interchange

- Support DMA dual address automatic switching

2.4.6. LED display

The TXW806 series chip integrates an LED display function module to expand LED display applications. Specific features

The properties are as follows:

- Support common cathode and common anode digital tubes

- Each COM and SEG has separate enablement

- Scan time is adjustable (two counters jointly control the scan time)

- Support interrupts

- Supports up to 8 com and 12 seg

- The display data register has a set of shadow registers, and you can choose whether to update the display data.

2.4.7. Camera DVP interface

The TXW806 series chip integrates a CMOS Sensor DVP interface function module for external expansion connections.

Camera. The specific features are as follows:

- Support Sensor 8 bits parallel interface;

- Support HSYNC and VSYNC polarity selection;

- Support HSYNC and VSYNC failure judgment;

- Supported data formats: YUV422, RGB565, JPEG and RAW DATA;

- Support YUV data sequence: YUYV, YVYU, UYVY and VYUY;

- Support adjusting sampling frame rate;

- Support single and continuous sampling modes;

- Supports retaining only Y data;

- Support YUV data range limitation;

- Support YUV and RGB image reduction by 1/2;

- Support YUV and RGB image cropping;

- Support color space conversion, RGB565 to YUV422;

- Support Scatter DMA, only YUV and RGB formats;

- Support Scatter DMA single/double Buffer mode;

2.4.8. Motion JPEG Video Encoder

The TXW806 series chip integrates a 1-channel MJPEG video encoder for compressing pictures or videos. Specific features

The properties are as follows:

- Support maximum resolution

- 1280*720 25fps

- 640*480 30fps

- Support pixel data format YUV420;

- Supports dual buffers for compressed data;

- Support DQT dynamic adjustment;

- Support generating JPEG header;

- Support DMA transmission;

2.4.9. Ethernet Controller MAC

The TXW806 series chip integrates a MAC function module for expanding Ethernet wired connections. specific characteristics

as follows:

- Support 10/100Mbps;

- Support RMII interface;

- Support half-duplex and full-duplex modes;

- Support up to 16 MAC address filtering;

- Support AMBA AHB 32bit;

- Support "ring" descriptor structure;

- Comply with IEEE 802.3 CSMA/CD standard.

2.4.10. SDIO2.0 Device Controller

The TXW806 series chip integrates an SDIO2.0 Device controller function module and supports SD 1/4bit mode.

mode, supports SDIO SPI mode; the clock supports up to 50MHz. For host SDIO the host interface can be

Conveniently connect and expand Wi-Fi applications.

2.4.11. SD HOST controller

The TXW806 series chip integrates an SD host controller function module and supports SD 1/4bit mode.

The clock supports up to 50MHz. Used to expand external SD memory cards or SDIO device interface devices.

2.4.12. M2M DMA module

The TXW806 series chip integrates 2 memory to memory DMA controller function modules. used in chip

DMA data transfer from SRAM to SRAM data stream. The specific features are as follows:

- Support byte alignment
- Supports up to 64KB transfer at a time
- Support memcpy and memset modes
- Supports 4 word cache buffers
- Support transfer completion flag and interrupt

2.5. Timer resources

2.5.1. Advanced timer

The TXW806 series chip integrates an advanced timer SUPTIMER.

The SUPTIMER module supports 6 PWM generators and can be configured into 6 independent PWM outputs.

(SUPTMR0-5), can also be configured as 3 pairs of complementary PWM (SUPTMR0-1,

SUPTMR2-3 and SUPTMR4-5).

Each PWM channel has a 10-bit prescaler. Each PWM output has an independent 16-bit counter for control

control, a 16-bit comparator is used to adjust the duty cycle. The 6-channel PWM generator provides 30 interrupt flags, and the related PWM pass

If the period or duty cycle of the track matches the counter, an interrupt flag will be generated. Each PWM channel has a separate enable bit.

Each PWM channel can be configured in single-shot mode (generating one PWM signal cycle) or cycle mode (continuously outputting PWM waves).

shape).

The SUPTIMER module has the following features:

- 6 independent 16-bit PWM control modes (SYNC synchronization, COMP complementary, GP group, default independent mode

- Mode);

- 6 independent PWM outputs: SUPTMR0-5;

- 3 sets of complementary PWM pair outputs: SUPTMR0-1, SUPTMR2-3, SUPTMR4-5;

- Programmable dead time can be inserted, 8 dead time modes are optional;

- 3 groups of synchronous PWM pair outputs: SUPTMR0-1, SUPTMR2-3, SUPTMR4-5;

- Support group control, SUPTMR0, SUPTMR2, SUPTMR4 output synchronization, SUPTMR1, SUPTMR3, SUPTMR5

- Output synchronization;

- Single mode or automatic loading mode;

- Supports edge alignment and center alignment modes; center alignment mode supports symmetric counting and asymmetric counting;

- Each PWM channel can independently select the output greater than the comparison value or less than the comparison value;

- Hardware brake protection (external FB trigger, comparator channel 0 or 1, ADC, supports software trigger); mutual

- Complement, synchronization, and group modes, please enable them first before configuring the period comparison value.

2.5.2. Basic timer

The TXW806 series chip integrates 4 Normal timers, of which timer0/3 is a 32-bit timer and timer1/2

It is a 16-bit timer with DMA and infrared functions.

The following only uses the 16-bit counter timer1 for explanation. The corresponding bit widths of timer0/3 are defined as 32 bits.

Timer timer1 consists of a 16-bit autoloader counter that can measure the pulse width of the input signal (input

capture), or generate output waveforms (output compare, PWM, etc.). Main features:

- 16-bit up counter

- Programmable (can be modified in real time) prescaler, supporting up to 128 times frequency division

- Supports selecting GPIO as counting clock source

- Support different counting clock sources

• Allows the timing period register to be updated after each counter cycle

• Support input capture function:

• Supports saving up to 1 capture event pointer at the same time

• The polarity of each capture event is independently configurable

• Each time a capture event occurs, you can configure whether to reset the count value.

• PWM output

• Support cycle and duty cycle auto-reload

• Use external signals to control timers and synchronization circuits interconnected by timers

• Support infrared function

2.5.3. Simple timer

The TXW806 series chip integrates a simple timer. The simple timer module is composed of timer0, Stimer1, Stimer2, Stimer3, composed of four 32-bit basic function timers, support a variety of counting clock source selections selection, supporting multiple working modes such as timer mode, counter mode, capture mode and PWM mode.

2.5.4. Watchdog timer

The TXW806 series chips integrate a watchdog module that is independent of system operation to protect the system from abnormal occurrences. A subsequent reset restarts the system. The working clock of the watchdog module is a normally open 64KHz low-speed RC divided-by-2 clock, which is Operating at 32KHz clock independent of the system clock. The default configuration is to reset the system every 2 seconds. So in the user program The watchdog needs to be fed before the watchdog is reset to retime it. The user can configure the watchdog reset interval ranging from 4ms ~ 1024s. The working mode can choose the watchdog to generate an interrupt or a direct reset.

2.6. Security hardware accelerator

2.6.1. CRC module

The TXW806 series chip integrates a CRC function module for data verification. The specific features are as follows:

- Support polynomials of different lengths such as 5/7/8/16/32 etc.

- Support custom polynomials

2.6.2. AES module

The TXW806 series chip integrates a system AES function module for data encryption and decryption. The specific features are as follows:

- Support AES-128/192/256 encryption and decryption

- Supports DMA mode, the maximum DMA data length is supported to 1MBytes

2.6.3. TRNG module

The TXW806 series chip integrates a true random number function module to generate random seeds for user data security.

3. Electrical parameters

3.1. Absolute maximum ratings (1)

symbol	parameter	Conditional	minimum value	maximum value	unit
V_{VCC}	working voltage		-0.3	3.6	IN
V_{VCCA}	simulates part of the operating voltage		-0.3	3.6	IN
V_{CCPA} V_{CCRF}	RF part working voltage		-0.3	3.6	IN
V_{CC1}	PB[6:15], PC[0:5] can pass V_{CC1} Set operating voltage domain		-0.3	3.6	IN
T_{ST}	storage temperature		-40	150	°C

(1) Operation outside absolute maximum rating conditions may cause permanent chip damage. The chip is working as recommended

Use under conditions other than those specified within the absolute maximum ratings may affect the chip's reliability, functionality, and performance, and shorten the chip's life.

3.2. Recommended working conditions

symbol	parameter	Condition	minimum value	typical value	maximum value	unit
V_{VCC}	working voltage		3	3.3	3.6	IN
V_{VCCA}	simulates part of the operating voltage		3	3.3	3.6	IN
V_{CCPA} V_{CCRF}	RF part working voltage		3	3.3	3.6	IN
V_{ICAM}	Vcam LDO load capacity	VCAM = 2.8V	56	112	140	mW
V_{ICAM}	Vcam LDO load capacity	VCAM = 1.8V	170	170	170	mW
V_{CC1}	PB[6:15], PC[0:5] can pass V_{CC1} Set the operating voltage domain (note: PA[0:15], PB[0:5] are fixed to VCC as voltage domain)		1.8		3.6	IN
T_A	working temperature		-40		85	°C

(1) The load capacity index of Vcam LDO is when the operating temperature is 85°C. Vcam LDO load capacity will become stronger as the temperature decreases;

It will become weaker as the VCC voltage decreases; it will become stronger as the VCAM voltage level decreases.

3.3. DC electrical characteristics

symbol	parameter	Minimum value	Typical value	Maximum value	Unit
CIN pin capacitance			2		pF
VIH high level input voltage		0.6*VCC		VCC+0.3	IN
VIL low level input voltage		-0.3		0.4*VCC	IN
IIH high level input current				50	nA
IIL low level input current				50	nA
VOH high level output voltage		0.9*VCC			IN
VOL2 low level output voltage				0.1*VCC	IN
John	High level source current (VCC = 3.3 V, VOH >= 2.64 V,PAD_DRIVER = 15)		60		mA
IOL	Low level sink current (VDD1 = 3.3 V, VOL = 0.495 V,PAD_DRIVER = 15)		60		mA
RPU pull-up resistor		4.7	10	100	kÿ
RPD pull-down resistor		4.7	10	100	kÿ
VIH_nRST chip MCLR reset release voltage		0.6*VCC		VCC+0.3	IN
VIL_nRST chip MCLR reset voltage		-0.3		0.4*VCC	IN
illustrate: 1. VCC and VCC1 are the power supplies of GPIO. PB[6:15], PC[0:5] can set the operating voltage domain through VCC1 (Note: PA[0:15], PB[0:5] is fixed to VCC operating voltage domain) 2. VOH and VOL are test values when the load is high resistance.					

3.4. AC electrical characteristics

3.4.1. External clock source characteristics

symbol	parameter	Condition	minimum value	typical value	maximum value	unit
f _{hoscm}	User external clock frequency			40		MHz

<i>VBIAS</i>	XOSCI/XOSCO bias level			550		mV
<i>Vxoh</i>	XOSCI input pin high level voltage-			0.77		mV
<i>Vxol</i>	XOSCO input pin low level voltage-			0.33		mV
Duty (<i>hoscm</i>)	duty cycle		42		58	%
<i>ACCxosc</i> HSE Accuracy						ppm
<i>t</i> <i>SU</i> (<i>xosc</i> m)	Start Time			5		ms
<i>IVCCA</i> , <i>XOSCM</i> ,	XOSCM oscillator power consumption	average power consumption		0.7		mA

3.4.2. Internal clock source characteristics

Table 2-3 RC10M oscillator characteristics

symbol	parameter	condition	Minimum value	Typical value	Maximum value	Unit
<i>VVCCA</i>	Supply voltage		3	3.3	3.6	V
RC10M	frequency	25y trim post test 9		10	11	MHz
<i>ACCRC10M</i>	RC10M oscillator accuracy -40y to 85y		-6		+6	%
<i>t</i> <i>SU</i> (<i>RC10M</i>)	RC10M oscillator start-up time -			60		us
<i>IVCCA</i> (<i>RC10M</i>)	RC10M oscillator power consumption	average power consumption			1	mA

Table 2-4 RC128K oscillator characteristics

symbol	parameter	Condition	minimum value	typical value	maximum value	unit
RC128K frequency		TA=25y		128		kHz
<i>ID</i> (<i>RC128K</i>)	RC128K Oscillator Power Consumption-			7.5		uA

3.5. Power consumption characteristics

3.5.1. RF power consumption

The following power consumption data are based on the test results of a 3.3V power supply, an ambient temperature of 25°C, and the CPU running 120MHz. All hair

The emission data are all measured based on 100% duty cycle.

RF Power consumption (measured at 100% duty cycle)			
Operating mode	describe		Mean(mA)
ActiveLDO Mode	TX	802.11g, 20 MHz, 54 Mbps, 15dBm	240
		802.11n, 20 MHz, MCS7, 15dBm	240
		802.11n, 20 MHz, MCS7, 6dBm	168
	RX	802.11b/g/n, 20 MHz	76

The following power consumption data are extrapolated:

RF Power consumption (50% duty cycle calculation)			
Operating mode	describe		Mean(mA)
ActiveLDO Mode TX+RX		802.11g, 20 MHz, 54 Mbps, 15dBm	158
		802.11n, 20 MHz, MCS7, 15dBm	158
		802.11n, 20 MHz, MCS7, 6dBm	122

RF Power consumption (measured at 100% duty cycle)			
Operating mode	describe		Mean(mA)
ActiveDCDC Mode	TX	802.11g, 20 MHz, 54 Mbps, 15dBm	216
		802.11n, 20 MHz, MCS7, 15dBm	216
		802.11n, 20 MHz, MCS7, 6dBm 802.11b/g/n,	152
	RX	20 MHz	43

The following power consumption figures are speculative:

RF Power consumption (50% duty cycle calculation)			
Operating mode	describe		Mean(mA)
ActiveDCDC Mode TX+RX		802.11g, 20 MHz, 54 Mbps, 15dBm	129
		802.11n, 20 MHz, MCS7, 15dBm	129
		802.11n, 20 MHz, MCS7, 6dBm	97

3.5.2. CPU power consumption

MCU status	WLAN status	TX/RX	Test Conditions		Power consumption
(1) LP	OFF		Normally-on power domain logic operation, 294KB		27.4 uA
			SRAM uninterruptible power supply		
	OFF		Normally open power domain logic operation, 32KB		8.7uA
			SRAM uninterruptible power supply		
ULP ⁽¹⁾	OFF				5.4 uA
Chip off			CHIP_EN is 0		0.7uA

(1) Turn off the internal RC and only support IO wake-up; if you turn on the internal RC wake-up, an additional 7.7uA power consumption is required

3.5.3. System sleep and keep-alive power consumption

illustrate:

The following power consumption data is based on built-in Buck DCDC Mode, 3.6V power supply, 25y ambient temperature, STA RX Beacon

The window is 1ms, tested in a shielded environment.

Test Conditions	DTIM1 (uA)	DTIM3 (uA)	DTIM5 (uA)	DTIM6 (uA)	DTIM10 (uA)	DTIM30 (uA)
External 32K	863	309	202	176	124	69
Built-in 32K	1073	414	281	246	177	96

Notice:

1. When using the internal 32K crystal oscillator, the longer the sleep time, the greater the error will be, so the power consumption will increase.
2. Wi-Fi PS Mode test power consumption will increase in interference environments, and power consumption data will vary in different interference environments.
3. The chip has a "D" built-in and supports Buck DCDC, such as TXW806-851D.

3.6. Reliability

3.6.1. ESD electrical characteristics

symbol	parameter	Test Conditions	Maximum unit level		
ESD	electrostatic discharge (Human Discharge Model HBM)	TA = +25°C, JEDEC EIA/JESD22-A114	6000	IN	
	electrostatic discharge (Component Charging Model CDM)	TA = +25°C, JEDEC EIA/JESD22-C101-B	1000	IN	

3.6.2. Latch-Up Electrical Characteristics

symbol	parameter	Test Conditions	Test type minimum value unit		
LU	Static latch-up class	JEDEC STANDARD NO.78D NOVEMBER 2011	Class I (TA = +25°C)	±200 mA	

3.7. Wi-Fi RF performance and power consumption

3.7.1. Transmitter performance

parameter	condition	Typical value(dBm)
Output Power	802.11g, 20 MHz, 54 Mbps	18.5
	802.11n, 20 MHz, MCS7	17.5

3.7.2. Receiver performance

parameter	condition	Typical value(dBm)
Receive sensitivity	HT20 MCS7 4k	-72
	NONHT 54M	-74.5

	NONHT 6M	-89.5
	CCK11M	-85
	CCK5.5M	-88
	DSSS2M	-91.5
	DSSS1M	-96

4. Reference Design

5. Ordering information

TXW806

850C

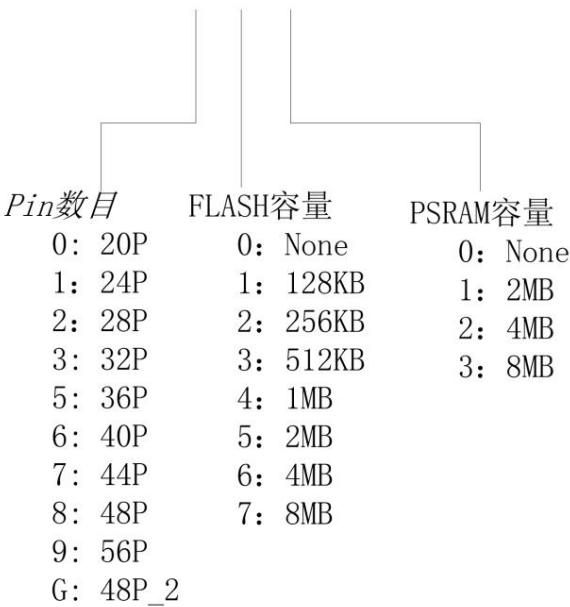


Figure 5-1 Model naming

Table 5-1 Ordering information

Product number	package size		describe
TXW806-850	QFN48	5x5 built-in	16Mbit Flash, video + audio application
TXW806-850U	QFN48	5x5	Built-in 16Mbit Flash, video + audio application, supports USB 2.0 HighSpeed Host
TXW806-851	QFN48	5x5	Built-in 16Mbit Flash + 16Mbit PSRAM, video + audio applications
TXW806-851D	QFN48	5x5	Built-in 16Mbit Flash + 16Mbit PSRAM, built-in BUCK DC-DC, video + audio low power application
TXW806-841	QFN48	5x5	Built-in 8Mbit Flash + 16Mbit PSRAM, video + audio frequency application
TXW806-840	QFN48	5x5 built-in	8Mbit Flash, video + audio application
TXW806-G00	QFN48	5x5 without	Flash\PSRAM, video + audio application